

Classwork Problems

3) Reduce the following matrices to Row reduced echelon form and determine their ranks. Identify the pivot variables & free variables. Find the special solutions to  $Ax=0$ .

i)  $\begin{bmatrix} 2 & 4 & -3 & -2 \\ -2 & -3 & 2 & -5 \\ 1 & 3 & -2 & 2 \end{bmatrix}$  ii)  $\begin{bmatrix} 0 & 2 & 3 & 7 \\ 1 & 1 & 1 & 1 \\ 1 & 3 & 4 & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Solution  
i)  $\begin{bmatrix} 2 & 4 & -3 & -2 \\ -2 & -3 & 2 & -5 \\ 1 & 3 & -2 & 2 \end{bmatrix} \xrightarrow[R_3 - R_1]{R_2 + R_1} \begin{bmatrix} 2 & 4 & -3 & -2 \\ 0 & 1 & -1 & -7 \\ 0 & 2 & -1 & 6 \end{bmatrix}$

$$\begin{bmatrix} 2 & 4 & -3 & -2 \\ 0 & 1 & -1 & -7 \\ 0 & 2 & -1 & 6 \end{bmatrix} \xrightarrow{R_3 - 2R_2} \begin{bmatrix} 2 & 4 & -3 & -2 \\ 0 & 1 & -1 & -7 \\ 0 & 0 & 1 & 20 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 4 & -3 & -2 \\ 0 & 1 & -1 & -7 \\ 0 & 0 & 1 & 20 \end{bmatrix} \xrightarrow{R_1/2} \begin{bmatrix} 1 & 2 & -3/2 & -1 \\ 0 & 1 & -1 & -7 \\ 0 & 0 & 1 & 20 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -3/2 & -1 \\ 0 & 1 & -1 & -7 \\ 0 & 0 & 1 & 20 \end{bmatrix} \xrightarrow{R_1 - 2R_2} \begin{bmatrix} 1 & 0 & 1/2 & 13 \\ 0 & 1 & -1 & -7 \\ 0 & 0 & 1 & 20 \end{bmatrix}$$

$$-\frac{3}{2} + \frac{2}{1} = \frac{-3+4}{2}$$

②

$$3 \text{ ii)} \begin{bmatrix} 0 & 2 & 3 & 7 \\ 1 & 1 & 1 & 1 \\ 1 & 3 & 4 & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_3 \leftrightarrow R_1} \begin{bmatrix} 1 & 3 & 4 & 8 \\ 1 & 1 & 1 & 1 \\ 0 & 2 & 3 & 7 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 4 & 8 \\ 1 & 1 & 1 & 1 \\ 0 & 2 & 3 & 7 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 - R_2} \begin{bmatrix} 1 & 3 & 4 & 8 \\ 0 & -2 & -3 & -7 \\ 0 & 2 & 3 & 7 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 4 & 8 \\ 0 & -2 & -3 & -7 \\ 0 & 2 & 3 & 7 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_3 + R_2} \begin{bmatrix} 1 & 3 & 4 & 8 \\ 0 & -2 & -3 & -7 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 4 & 8 \\ 0 & -2 & -3 & -7 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_4 \leftrightarrow R_3} \begin{bmatrix} 1 & 3 & 4 & 8 \\ 0 & -2 & -3 & -7 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(3)

$$\begin{bmatrix} 1 & 3 & 4 & 8 \\ 0 & -2 & -3 & -7 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 \rightarrow \frac{R_2}{-2}} \begin{bmatrix} 1 & 3 & 4 & 8 \\ 0 & 1 & 3/2 & 7/2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 4 & 8 \\ 0 & 1 & 3/2 & 7/2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 - 3R_2} \begin{bmatrix} \textcircled{1} & 0 & -1/2 & -5/2 \\ 0 & \textcircled{1} & 3/2 & 7/2 \\ 0 & 0 & 0 & \textcircled{1} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$4 - 9/2 \quad 8 - 21/2$$

Pivot variables =  $x, y, t$ ; Free variables =  $z$   
 special solutions  $Rx = 0$

$$t = 0; \quad y + 3/2 z + 7/2 t = 0 \Rightarrow y + 3/2 z = 0$$

$$x - z/2 - 5/2 t = 0 \Rightarrow x - z/2 = 0 \Rightarrow \boxed{x = z/2}$$

$$y = -3/2 z$$

$$\begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} z/2 \\ -3/2 z \\ z \\ 0 \end{bmatrix} = z \begin{bmatrix} 1/2 \\ -3/2 \\ 1 \\ 0 \end{bmatrix}$$

(4)

1 i) Let  $v_1 = (1, 2, 3)$  and  $v_2 = (4, 5, 6)$  and  $v_3 = (2, 1, 0)$

a) Determine if the set  $(v_1, v_2, v_3)$  is linearly independent?

Solution :-

$$\begin{bmatrix} 1 & 4 & 2 \\ 2 & 5 & 1 \\ 3 & 6 & 0 \end{bmatrix} \xrightarrow{\substack{R_2 - 2R_1 \\ R_3 - 3R_1}} \begin{bmatrix} 1 & 4 & 2 \\ 0 & -3 & -3 \\ 0 & -6 & -6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4 & 2 \\ 0 & -3 & -3 \\ 0 & -6 & -6 \end{bmatrix} \xrightarrow{R_3 - 2R_2} \begin{bmatrix} 1 & 4 & 2 \\ 0 & -3 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

Since  $\rho(A) = 2 < n \Rightarrow$  Given set of vectors are linearly dependent.

b) If, possible find a linear dependence relation among  $v_1, v_2, v_3$ ? Infinite solutions

$$-3c_2 - 3c_3 = 0 \quad c_1 + 4c_2 + 2c_3 = 0$$

$$-3c_2 = 3c_3$$

$$c_2 = -c_3$$

$$\text{let } c_3 = K$$

$$c_1 + 4(-K) + 2K = 0$$

$$c_1 - 2K = 0 \quad c_1 = 2K$$

If  $K = 5$

$$10v_1 - 5v_2 + 5v_3 = 0$$

1 ii) Find the value of 'h' for which the vectors  $(1, -1, 4)$ ,  $(3, 5, -7)$  and  $(-1, 5, h)$  are dependent?

Solution

$$\begin{bmatrix} 1 & 3 & -1 \\ -1 & 5 & 5 \\ 4 & -7 & h \end{bmatrix} \xrightarrow{\substack{R_2 + R_1 \\ R_3 - 4R_1}} \begin{bmatrix} 1 & 3 & -1 \\ 0 & 8 & 4 \\ 0 & -19 & h+4 \end{bmatrix}$$

$$-19 - 8x = 0$$

$$-19 = 8x$$

$$x = -\frac{19}{8}$$

$$\begin{bmatrix} 1 & 3 & -1 \\ 0 & 8 & 4 \\ 0 & -19 & h+4 \end{bmatrix} \xrightarrow{R_3 + \frac{19}{8}R_2} \begin{bmatrix} 1 & 3 & -1 \\ 0 & 8 & 4 \\ 0 & 0 & \frac{2h+27}{2} \end{bmatrix}$$

$$h+4 + \frac{19 \times 4}{8} = 1$$

$$\begin{aligned} h+4 + \frac{19}{2} \\ \frac{1}{2}h + 8 + \frac{19}{2} \end{aligned}$$

11) Let  $V$  be a subspace of the 4-dimensional vector space  $\mathbb{R}^4$  (5)

$$V = \left\{ \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} \in \mathbb{R}^4 \mid a - b + c - d = 0 \right\}. \text{ Find the basis \& dimension of } V?$$

Solution

$$\begin{bmatrix} b - c + d \\ b \\ c \\ d \end{bmatrix} = b \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + d \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

∴ Basis vectors =  $\begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

Dimension = 3

1 iii) Examine if the following sets of vectors are linearly independent. When the set is dependent, find a relation between the vectors.

a)  $\{t^2 - t + 5, 2t^2 - 3t, -t^2 + 2t + 5\}$

$$A = \begin{bmatrix} 1 & 2 & -1 \\ -1 & -3 & 2 \\ 5 & 0 & 5 \end{bmatrix} \xrightarrow[R_3 - 5R_1]{R_2 + R_1} \begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 1 \\ 0 & -10 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 1 \\ 0 & -10 & 10 \end{bmatrix} \xrightarrow{R_3 - 10R_2} \begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

The given set of vectors are linearly dependent.

$$c_1 + 2c_2 - c_3 = 0 \quad -c_2 + c_3 = 0$$

$$\text{let } c_3 = k \quad -c_2 = -c_3$$

$$c_1 + 2c_2 - c_3 = 0$$

$$c_1 + 2k - k = 0 \quad c_1 + k = 0$$

$$c_2 = c_3$$

$$c_1 = -k$$

⑥

$$b) \left\{ \begin{pmatrix} 1 & -5 \\ -4 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ -1 & 5 \end{pmatrix}, \begin{pmatrix} -2 & 3 \\ 5 & 2 \end{pmatrix}, \begin{pmatrix} 1 & -2 \\ -5 & 3 \end{pmatrix} \right\} \text{ in } M_{2 \times 2}(\mathbb{R})$$

Solution

$$A = \begin{bmatrix} 1 & 1 & -2 & 1 \\ -4 & -1 & 5 & -5 \\ -5 & 1 & 3 & -2 \\ 2 & 5 & 2 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & -2 & 1 \\ -4 & -1 & 5 & -5 \\ -5 & 1 & 3 & -2 \\ 2 & 5 & 2 & 3 \end{bmatrix} \xrightarrow{\substack{R_2 + 4R_1 \\ R_3 + 5R_1 \\ R_4 - 2R_1}} \begin{bmatrix} 1 & 1 & -2 & 1 \\ 0 & 3 & -3 & -1 \\ 0 & 6 & -7 & 3 \\ 0 & 3 & 6 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & -2 & 1 \\ 0 & 3 & -3 & -1 \\ 0 & 6 & -7 & 3 \\ 0 & 3 & 6 & 3 \end{bmatrix} \xrightarrow{\substack{R_3 - 2R_2 \\ R_4 - R_2}} \begin{bmatrix} 1 & 1 & -2 & 1 \\ 0 & 3 & -3 & -1 \\ 0 & 0 & -1 & 5 \\ 0 & 0 & 12 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & -2 & 1 \\ 0 & 3 & -3 & -1 \\ 0 & 0 & -1 & 5 \\ 0 & 0 & 12 & 5 \end{bmatrix} \xrightarrow{R_4 + 12R_3} \begin{bmatrix} ① & 1 & -2 & 1 \\ 0 & ③ & -3 & -1 \\ 0 & 0 & ① & 5 \\ 0 & 0 & 0 & ⑥⑤ \end{bmatrix}$$

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Since  $\rho(A) = 4 = n$ , given set of vectors are linearly independent.

5. Let  $W$  be the set of vectors of the form

$\begin{bmatrix} 5b+2c \\ b \\ c \end{bmatrix}$ , where  $b$  &  $c$  are arbitrary. Find the vectors  $u$  &  $v$  such that  $W = \text{span}(u, v)$ . Why does this show that  $W$  is a subspace of  $\mathbb{R}^3$ .

Solution :-

$$\begin{bmatrix} 5b+2c \\ b \\ c \end{bmatrix} = b \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

$$\therefore u = \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix} \quad v = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

If  $v_1, v_2, \dots, v_p$  are in vector space  $V$ , then  $\text{span}\{v_1, v_2, \dots, v_p\}$  is a subspace of  $V$ , as is a subspace of  $\mathbb{R}^3$ .

8. i) For which vectors  $b = (a, b, c)$  do the following system  $Ax = b$  have a solution?

$$x+2y=a; -x+y+2z=b; 3x-4z=c$$

Solution :-

$$\begin{bmatrix} 1 & 2 & 0 & : & a \\ -1 & 1 & 2 & : & b \\ 3 & 0 & -4 & : & c \end{bmatrix} \xrightarrow[R_3-3R_1]{R_2+R_1} \begin{bmatrix} 1 & 2 & 0 & : & a \\ 0 & 3 & 2 & : & b+a \\ 0 & -6 & -4 & : & c-3a \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 0 & : & a \\ 0 & 3 & 2 & : & b+a \\ 0 & -6 & -4 & : & c-3a \end{bmatrix} \xrightarrow{R_3+2R_2} \begin{bmatrix} 1 & 2 & 0 & : & a \\ 0 & 3 & 2 & : & b+a \\ 0 & 0 & 0 & : & c-3a+2b+2a \end{bmatrix}$$

For the system of equations to have a solution,

$$\boxed{c+2b-a=0}$$

6. If  $A = \begin{bmatrix} 1 & -1 & 5 \\ 2 & 0 & 7 \\ -3 & -5 & -3 \end{bmatrix}$

$$\xrightarrow[R_3+3R_1]{R_2-2R_1} \begin{bmatrix} 1 & -1 & 5 \\ 0 & 2 & -3 \\ 0 & -8 & 12 \end{bmatrix}$$

$$\xrightarrow{R_3+4R_2} \begin{bmatrix} 1 & -1 & 5 \\ 0 & 2 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

and  $u = \begin{pmatrix} -7 \\ 3 \\ 2 \end{pmatrix}$ . Is  $u$  in nullspace of  $A$ ? Is  $u$  in column space of  $A$ ? Justify your answer.

8

$$\begin{bmatrix} 1 & -1 & 5 \\ 0 & 2 & -3 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$2y - 3z = 0$$

$$2y = 3z$$

$$y = \frac{3}{2}z ; \text{ let } z=2 \Rightarrow y=3$$

$$x - y + 5z = 0$$

$$x - 3 + 10 = 0$$

$$x = -7$$

∴  $u = \begin{pmatrix} -7 \\ 3 \\ 2 \end{pmatrix}$  is in null space of  $A$ .

The system of equations  $Ax=b$  is solvable iff the vector  $b$  can be expressed as a combination of columns of  $A$ . Then  $b$  is in column space of  $A$ .

$$\text{Here } \alpha \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} + \beta \begin{bmatrix} -1 \\ 0 \\ -5 \end{bmatrix} + \gamma \begin{bmatrix} 5 \\ 7 \\ -3 \end{bmatrix} = \begin{bmatrix} -7 \\ 3 \\ 2 \end{bmatrix}$$

for  $u = \begin{bmatrix} -7 \\ 3 \\ 2 \end{bmatrix}$  to be in  $C(A)$

$$\text{If } \alpha = -1; \beta = 1; \gamma = -1 ; \begin{aligned} -1 - 1 - 5 &= -7 \\ -2 - 7 &= -9 \end{aligned}$$

$$3 - 1 + 3 - 5 = 1$$

∴  $u$  is not in column space of  $A$ .

2) (a) Determine whether these vectors

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$(1, 1, 1, 1)$   $(1, 2, 3, 2)$ ,  $(2, 5, 6, 4)$   $(2, 6, 8, 5)$  form a basis for  $\mathbb{R}^4$ . If not, find the dimension of the subspaces  $S$  they span.

(b) If  $S$  is a subspace of  $\mathbb{R}^4$ , extend the basis of  $S$  to a basis of  $\mathbb{R}^4$ .

Solution (a)  $A = \begin{bmatrix} 1 & 1 & 2 & 2 \\ 1 & 2 & 5 & 6 \\ 1 & 3 & 6 & 8 \\ 1 & 2 & 4 & 5 \end{bmatrix} \xrightarrow[\substack{R_2 - R_1 \\ R_3 - R_1 \\ R_4 - R_1}]{\quad} \begin{bmatrix} 1 & 1 & 2 & 2 \\ 0 & 1 & 3 & 4 \\ 0 & 2 & 4 & 6 \\ 0 & 1 & 2 & 3 \end{bmatrix}$

$$\begin{bmatrix} 1 & 1 & 2 & 2 \\ 0 & 1 & 3 & 4 \\ 0 & 2 & 4 & 6 \\ 0 & 1 & 2 & 3 \end{bmatrix} \xrightarrow[\substack{R_3 - 2R_2 \\ R_4 - R_2}]{\quad} \begin{bmatrix} 1 & 1 & 2 & 2 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & -1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 2 & 2 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & -1 & -1 \end{bmatrix} \xrightarrow{R_4 - \frac{1}{2}R_3} \begin{bmatrix} \textcircled{1} & 1 & 2 & 2 \\ 0 & \textcircled{1} & 3 & 4 \\ 0 & 0 & \textcircled{-2} & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

They do not form a basis of  $\mathbb{R}^4$ . They span a subspace  $S$  of dimension 3.

10

$$\begin{bmatrix} 1 & 1 & 2 & 2 & 1 & 0 & 0 & 0 \\ 1 & 2 & 5 & 6 & 0 & 1 & 0 & 0 \\ 1 & 3 & 6 & 8 & 0 & 0 & 1 & 0 \\ 1 & 2 & 4 & 5 & 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\substack{R_2 - R_1 \\ R_3 - R_1 \\ R_4 - R_1}} \begin{bmatrix} 1 & 1 & 2 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 3 & 4 & -1 & 1 & 0 & 0 \\ 0 & 2 & 4 & 6 & -1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 3 & -1 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 2 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 3 & 4 & -1 & 1 & 0 & 0 \\ 0 & 2 & 4 & 6 & -1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 3 & -1 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\substack{R_3 - 2R_2 \\ R_4 - R_2}} \begin{bmatrix} 1 & 1 & 2 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 3 & 4 & -1 & 1 & 0 & 0 \\ 0 & 0 & -2 & -2 & 1 & -2 & 1 & 0 \\ 0 & 0 & -1 & -1 & 0 & -1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 2 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 3 & 4 & -1 & 1 & 0 & 0 \\ 0 & 0 & -2 & -2 & 1 & -2 & 1 & 0 \\ 0 & 0 & -1 & -1 & 0 & -1 & 0 & 1 \end{bmatrix} \xrightarrow{R_4 - \frac{1}{2}R_3} \begin{bmatrix} \textcircled{1} & 1 & 2 & 2 & 1 & 0 & 0 & 0 \\ 0 & \textcircled{1} & 3 & 4 & -1 & 1 & 0 & 0 \\ 0 & 0 & \textcircled{-2} & -2 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 & \textcircled{-\frac{1}{2}} & \textcircled{-\frac{1}{2}} & \textcircled{\frac{1}{2}} & \textcircled{\frac{1}{2}} \end{bmatrix}$$

∴ Basis vectors :-  $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 2 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \\ 6 \\ 4 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

12) i) Find a subset of the vectors that form a basis for the space spanned by the vectors, then express each vector that is not in the basis as a linear combination of the basis vectors.

$v_1 = (1, 0, 1, 1); v_2 = (-4, 4, -1, 4); v_3 = (2, 4, 5, 10); v_4 = (-10, 4, -7, 2)$   
Solution

$$\begin{bmatrix} 1 & -4 & 2 & -10 \\ 0 & 4 & 4 & 4 \\ 1 & -1 & 5 & -7 \\ 1 & 4 & 10 & 2 \end{bmatrix} \xrightarrow[\substack{R_3 - R_1 \\ R_4 - R_1}]{R_3 - R_1} \begin{bmatrix} 1 & -4 & 2 & -10 \\ 0 & 4 & 4 & 4 \\ 0 & 3 & 3 & 3 \\ 0 & 8 & 8 & 12 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -4 & 2 & -10 \\ 0 & 4 & 4 & 4 \\ 0 & 3 & 3 & 3 \\ 0 & 8 & 8 & 12 \end{bmatrix} \xrightarrow[\substack{R_3 - \frac{3}{4}R_2 \\ R_4 - 2R_2}]{R_3 - \frac{3}{4}R_2} \begin{bmatrix} 1 & -4 & 2 & -10 \\ 0 & 4 & 4 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -4 & 2 & -10 \\ 0 & 4 & 4 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix} \xrightarrow{R_3 \leftrightarrow R_4} \begin{bmatrix} \textcircled{1} & -4 & 2 & -10 \\ 0 & \textcircled{4} & 4 & 4 \\ 0 & 0 & 0 & \textcircled{4} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\therefore \{v_1, v_2, v_4\}$  spans subspace of  $\mathbb{R}^4$ .

$$c_1 \cdot v_1 + c_2 \cdot v_2 + c_3 \cdot v_3 = v_3$$

(12)

$$\begin{bmatrix} 1 & -4 & -10 & : & 2 \\ 0 & 4 & 4 & : & 4 \\ 1 & -1 & -7 & : & 5 \\ 1 & 4 & 2 & : & 10 \end{bmatrix} \xrightarrow{\substack{R_3 - R_1 \\ R_4 - R_1}} \begin{bmatrix} 1 & -4 & -10 & : & 2 \\ 0 & 4 & 4 & : & 4 \\ 0 & 3 & 3 & : & 3 \\ 0 & 8 & 12 & : & 8 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -4 & -10 & : & 2 \\ 0 & 4 & 4 & : & 4 \\ 0 & 3 & 3 & : & 3 \\ 0 & 8 & 12 & : & 8 \end{bmatrix} \xrightarrow{\substack{R_3 - \frac{3}{4}R_2 \\ R_4 - 2R_2}} \begin{bmatrix} 1 & -4 & -10 & : & 2 \\ 0 & 4 & 4 & : & 4 \\ 0 & 0 & 0 & : & 0 \\ 0 & 0 & 4 & : & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -4 & -10 & : & 2 \\ 0 & 4 & 4 & : & 4 \\ 0 & 0 & 0 & : & 0 \\ 0 & 0 & 4 & : & 0 \end{bmatrix} \xrightarrow{R_4 \leftrightarrow R_3} \begin{bmatrix} \textcircled{1} & -4 & -10 & : & 2 \\ 0 & \textcircled{4} & 4 & : & 4 \\ 0 & 0 & \textcircled{4} & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

$$4c_3 = 0 \Rightarrow \boxed{c_3 = 0} \quad 4c_2 + 4c_3 = 4$$

$$4c_2 = 4$$

$$\boxed{c_2 = 1}$$

$$c_1 - 4c_2 - 10c_3 = 2$$

$$c_1 - 4 = 2$$

$$\boxed{c_1 = 6}$$

$$\therefore \boxed{v_3 = 6 \cdot v_1 + v_2}$$

12 ii) Find the conditions  $(a, b, c)$  such that  $a, b, c \in \text{span}(2, 1, 0), (1, -1, 2), (0, 3, -4)$ . Do the vectors span  $\mathbb{R}^3$ .

Solution

$$\left[ \begin{array}{ccc|c} 2 & 1 & 0 & a \\ 1 & -1 & 3 & b \\ 0 & 2 & -4 & c \end{array} \right] \xrightarrow{R_2 - \frac{1}{2}R_1} \left[ \begin{array}{ccc|c} 2 & 1 & 0 & a \\ 0 & -3/2 & 3 & b - a/2 \\ 0 & 2 & -4 & c \end{array} \right]$$

$$-1 - 1/2$$

$$2 - x(-3/2) = 0$$

$$2 + \frac{3x}{2} = 0$$

$$2 = -\frac{3x}{2}$$

$$x = -\frac{4}{3}$$

$$\left[ \begin{array}{ccc|c} 2 & 1 & 0 & a \\ 0 & -3/2 & 3 & \frac{2b-a}{2} \\ 0 & 2 & -4 & c \end{array} \right] \xrightarrow{R_3 + \frac{4}{3}R_2} \left[ \begin{array}{ccc|c} 2 & 1 & 0 & a \\ 0 & (-3/2) & 3 & \frac{2b-a}{2} \\ 0 & 0 & 0 & \frac{3c+4b-2a}{3} \end{array} \right]$$

$$-4 + \frac{4}{3}(3) = 0 \quad c + \frac{4}{3} \left\{ \frac{2b-a}{2} \right\}$$

$$= c + \frac{4b-2a}{3} = \frac{3c+4b-2a}{3}$$

Condition on  $a, b, c$  is  $2a - 4b - 3c = 0$

13) If the column space of  $A$  is spanned by the vectors  $(1, 0, -1), (2, 1, 3), (4, 2, 6), (3, 1, 2)$ , find all those vectors that span the null space of  $A$ . Determine whether or not the vector  $b = (-2, -2, 0, 2)$  is in that subspace. What are the bases and dimensions of  $C(A^T)$  &  $N(A^T)$ .

(14)

Solution TO find all those vectors that span the null space of A, we have to find the special solutions :-

$$\begin{bmatrix} 1 & 2 & 4 & 3 \\ 0 & 1 & 2 & 1 \\ -1 & 3 & 6 & 2 \end{bmatrix} \xrightarrow{R_3 + R_1} \begin{bmatrix} 1 & 2 & 4 & 3 \\ 0 & 1 & 2 & 1 \\ 0 & 5 & 10 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 4 & 3 \\ 0 & 1 & 2 & 1 \\ 0 & 5 & 10 & 5 \end{bmatrix} \xrightarrow{R_3 - 5R_2} \begin{bmatrix} 1 & 2 & 4 & 3 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 4 & 3 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$x, y$  = pivot variables ;  $z, t$  = free variables

$$y + 2z + t = 0 \Rightarrow y = -2z - t$$

$$x + 2y + 4z + 3t = 0 \Rightarrow x = -2y - 4z - 3t$$

$$\text{let } y = 1, z = 0, t = 1; y = -1; x = 2 - 3 = -1$$

special solutions  $\begin{bmatrix} -1 \\ 1 \\ 0 \\ 1 \end{bmatrix}$

$$\text{let } z = 1, t = 0; y = -2; x = 4 - 4 = 0$$

special solutions  $\begin{bmatrix} 0 \\ -2 \\ 1 \\ 0 \end{bmatrix}$

To determine whether or not the vector

$b = (-2, -2, 0, 2)$  is in the nullspace, we have to

find  $A \cdot b = 0$

$$\begin{bmatrix} 1 & 2 & 4 & 3 \\ 0 & 1 & 2 & 1 \\ -1 & 3 & 6 & 2 \end{bmatrix} \begin{bmatrix} -2 \\ -2 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 - 4 + 0 + 6 \\ 0 - 2 + 0 + 2 \\ 2 - 6 + 0 + 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

∴ Vector  $b$  is in nullspace of  $A$ .

To find the bases & dimensions of  $C(A)$  &  $N(A)$  ∴

$$C(A) = \left\{ c_1 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} \mid c_1, c_2 \in \mathbb{R} \right\}$$

$$C(A^T) = \left\{ c_1 \begin{bmatrix} 1 \\ 2 \\ 4 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 2 \\ 1 \end{bmatrix} \mid c_1, c_2 \in \mathbb{R} \right\}$$

$$\dim \{C(A^T)\} = 2$$

$$A = \begin{bmatrix} 1 & 2 & 4 & 3 \\ 0 & 1 & 2 & 1 \\ -1 & 3 & 6 & 2 \end{bmatrix}; A^T = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 3 \\ 4 & 2 & 6 \\ 3 & 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 3 \\ 4 & 2 & 6 \\ 3 & 1 & 2 \end{bmatrix} \xrightarrow[\substack{R_2 - 2R_1 \\ R_3 - 4R_1 \\ R_4 - 3R_1}]{\rightarrow} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 5 \\ 0 & 2 & 10 \\ 0 & 1 & 5 \end{bmatrix}$$

(16)

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 5 \\ 0 & 2 & 10 \\ 0 & 1 & 5 \end{bmatrix} \xrightarrow[R_4 - R_2]{R_3 - 2R_2} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$x, y$  = pivot variables ;  $z$  = free variable

$$y + 5z = 0 ; y = -5z$$

$$x - z = 0 ; x = z$$

special solution

$$\begin{bmatrix} 1 \\ -5 \\ 1 \end{bmatrix}$$

$$\dim\{N(AT)\} = 1$$

4) Find the column space and Null space for the following matrices :-

i)  $\begin{bmatrix} 2 & 4 & -2 & 2 \\ -2 & 5 & 7 & 3 \\ -3 & 6 & -8 & 6 \end{bmatrix}$

ii)  $\begin{bmatrix} -2 & 1 & 2 & 1 \\ 0 & 3 & -1 & -2 \\ 2 & -3 & 0 & 4 \\ 1 & -1 & 2 & -2 \end{bmatrix}$

Solution :- i)  $\begin{bmatrix} 2 & 4 & -2 & 2 \\ -2 & 5 & 7 & 3 \\ -3 & 6 & -8 & 6 \end{bmatrix} \xrightarrow[R_3 + \frac{3}{2}R_1]{R_2 + R_1} \begin{bmatrix} 2 & 4 & -2 & 2 \\ 0 & 9 & 5 & 5 \\ 0 & 12 & -11 & 9 \end{bmatrix}$

$$\begin{aligned} -3 - 2x &= 0 & 6 + \frac{3}{2}(4) &= 12 & 6 + \frac{3}{2}(2) \\ -3 &= 2x & & & \\ \boxed{x = -3/2} & & -8 + \frac{3}{2}(-2) & & \end{aligned}$$

$$\begin{bmatrix} 2 & 4 & -2 & 2 \\ 0 & 9 & 5 & 5 \\ 0 & 12 & -11 & 9 \end{bmatrix} \xrightarrow{R_3 - \frac{4}{3}R_2} \begin{bmatrix} 2 & 4 & -2 & 2 \\ 0 & 9 & 5 & 5 \\ 0 & 0 & -\frac{53}{3} & \frac{7}{3} \end{bmatrix}$$

$$\begin{aligned} 12 - 9x &= 0 \Rightarrow 12 = 9x & -11 - \frac{4}{3}(5) &= \frac{-11 - \frac{20}{3}}{1} \\ 9 - \frac{4}{3}(5) &= \frac{9 - \frac{20}{3}}{1} & \boxed{x = 4/3} & ; & = \frac{-33 - 20}{3} \end{aligned}$$

$$C(A) = c_1 \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} + c_3 \begin{bmatrix} -2 \\ 7 \\ -8 \end{bmatrix}$$

$C(A)$  is a 3D plane in  $\mathbb{R}^3$ .

TO find  $N(A)$  :-

$$\begin{bmatrix} 2 & 4 & -2 & 2 \\ 0 & 9 & 5 & 5 \\ 0 & 0 & -\frac{53}{3} & \frac{7}{3} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$x, y, z$  = pivot variables ;  $t$  = free variable

$$-\frac{53}{3}z + \frac{7}{3}t = 0 ; -53z = -7t$$

$$\boxed{z = \frac{7}{53}} \text{ if } t = 1$$

$$9y + 5z + 5t = 0$$

$$9y + \frac{35}{53} + 5 = 0$$

$$\frac{53 \times 5}{265}$$

$$\frac{1}{265} \frac{35}{00}$$

$$9y = -\frac{5}{1} - \frac{35}{53}$$

$$9y = \frac{-265 - 35}{53} = \frac{-300}{53}$$

$$9y = \frac{-300}{53} \Rightarrow y = \frac{-300}{53 \times 9} = \frac{-100}{53 \times 3} = \frac{-100}{159}$$

$$\boxed{y = \frac{-100}{159}}$$

$$2x + 4y - 2z + 2t = 0$$

$$2x - \frac{400}{159} - \frac{14}{53} + 2 = 0$$

$$2x = \frac{400}{159} + \frac{14}{53} - \frac{2}{1}$$

$$2x = \frac{400 + 42 - 318}{159} = \frac{124}{159}$$

$$\begin{bmatrix} \frac{66}{159} \\ \frac{-100}{159} \\ \frac{7}{53} \\ 1 \end{bmatrix}$$

null space is a line in  $\mathbb{R}^4$ .

$$\boxed{x = \frac{66}{159}}$$

$$\begin{array}{r} 442 \\ 318 \\ \hline 124 \end{array}$$

$$\frac{53 \times 3}{159}$$

$$\frac{1}{159}$$

$$\frac{159}{18}$$

(18)

$$4 \text{ eqs)} \begin{bmatrix} -2 & 1 & 2 & 1 \\ 0 & 3 & -1 & -2 \\ 2 & -3 & 0 & 4 \\ 1 & -1 & 2 & -2 \end{bmatrix} \xrightarrow[\substack{R_3 + R_1 \\ R_4 + \frac{1}{2}R_1}]{\substack{R_3 + R_1 \\ R_4 + \frac{1}{2}R_1}} \begin{bmatrix} -2 & 1 & 2 & 1 \\ 0 & 3 & -1 & -2 \\ 0 & -2 & 2 & 5 \\ 0 & -1/2 & 3 & -3/2 \end{bmatrix}$$

$$-2 + 1/2$$

$$\begin{bmatrix} -2 & 1 & 2 & 1 \\ 0 & 3 & -1 & -2 \\ 0 & -2 & 2 & 5 \\ 0 & -1/2 & 3 & -3/2 \end{bmatrix} \xrightarrow[\substack{R_3 + 2/3 R_2 \\ R_4 + 1/6 R_2}]{\substack{R_3 + 2/3 R_2 \\ R_4 + 1/6 R_2}} \begin{bmatrix} -2 & 1 & 2 & 1 \\ 0 & 3 & -1 & -2 \\ 0 & 0 & 4/3 & 11/3 \\ 0 & 0 & 17/6 & -11/6 \end{bmatrix}$$

$$-2 - 3x = 0$$

$$-2 = 3x$$

$$\boxed{x = -2/3}$$

$$-1/2 - 3x = 0$$

$$-1/2 = 3x$$

$$\boxed{x = -1/6}$$

$$3 - 1/6$$

$$\frac{1}{18 - 1}$$

$$6$$

$$\frac{-3 - 1}{2 \quad 3}$$

$$6$$

$$\frac{17 \times 11}{17}$$

$$\frac{17 \times}{187}$$

$$\frac{2 - 2}{1 \quad 6 - 2 \quad 3}$$

$$3$$

$$\frac{5 - 4}{1 \quad 3}$$

$$3$$

$$\frac{17}{6} - \frac{4}{3}x = 0$$

$$\frac{17}{6} = \frac{4}{3}x$$

$$x = \frac{17 \times 3}{4 \times 6} = \frac{17}{8}$$

$$\begin{bmatrix} -2 & 1 & 2 & 1 \\ 0 & 3 & -1 & -2 \\ 0 & 0 & 4/3 & 11/3 \\ 0 & 0 & 17/6 & -11/6 \end{bmatrix} \xrightarrow{R_3 - \frac{17}{8}R_2} \begin{bmatrix} -2 & 1 & 2 & 1 \\ 0 & 3 & -1 & -2 \\ 0 & 0 & 4/3 & 11/3 \\ 0 & 0 & 0 & -231/4 \end{bmatrix}$$

$C(A)$  is a 4-D plane  
in  $\mathbb{R}^4$ .

$$-\frac{11}{6} - \frac{17}{8} \frac{11}{3} = -\frac{11}{6} - \frac{187}{24}$$

$$= \frac{-44 - 187}{24} = \frac{-231}{24}$$

4 ii) To find the null space  $N(A)$  :-

$$\begin{bmatrix} -2 & 1 & 2 & 1 \\ 0 & 3 & -1 & -2 \\ 0 & 0 & 4/3 & 11/3 \\ 0 & 0 & 0 & -231/24 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-\frac{231}{24}t = 0 \Rightarrow t = 0$$

$$\frac{4}{3}z + \frac{11}{3}t = 0 \Rightarrow z = 0$$

$$3y - z - 2t = 0 \Rightarrow y = 0$$

$$-2x + y + 2z + t = 0 \Rightarrow x = 0$$

$$\therefore \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \cdot N(A) \text{ is the origin in } \mathbb{R}^4.$$

14) Obtain the four fundamental subspaces, their bases and dimension given

$$\begin{bmatrix} 1 & 1 & -2 & 0 & -1 \\ 1 & -1 & 2 & -3 & 1 \\ 2 & 2 & -1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

Solution

$$\begin{bmatrix} 1 & 1 & -2 & 0 & -1 \\ 1 & -1 & 2 & -3 & 1 \\ 2 & 2 & -1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix} \xrightarrow[R_2 - 2R_1]{R_2 - R_1} \begin{bmatrix} 1 & 1 & -2 & 0 & -1 \\ 0 & -2 & 4 & -3 & 2 \\ 0 & 0 & 3 & 0 & 3 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & -2 & 0 & -1 \\ 0 & -2 & 4 & -3 & 2 \\ 0 & 0 & 3 & 0 & 3 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix} \xrightarrow{R_4 - \frac{1}{3}R_3} \begin{bmatrix} \textcircled{1} & 1 & -2 & 0 & -1 \\ 0 & \textcircled{-2} & 4 & -3 & 2 \\ 0 & 0 & \textcircled{3} & 0 & 3 \\ 0 & 0 & 0 & \textcircled{1} & 0 \end{bmatrix}$$

(20)

$$C(A) = c_1 \begin{bmatrix} 1 \\ 1 \\ 2 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -1 \\ 2 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} -2 \\ 2 \\ -1 \\ 1 \end{bmatrix} + c_4 \begin{bmatrix} 0 \\ -3 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{Dimension } \{C(A)\} = 4$$

$$C(AT) = c_1 \begin{bmatrix} 1 \\ 1 \\ -2 \\ 0 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -1 \\ 2 \\ 0-3 \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} +2 \\ 2 \\ -1 \\ 0 \\ 1 \end{bmatrix} + c_4 \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\text{Dimension } \{C(AT)\} = 4$$

To find  $N(A)$  :-

$$\begin{bmatrix} \textcircled{1} & 1 & -2 & 0 & -1 \\ 0 & \textcircled{-2} & 4 & -3 & 2 \\ 0 & 0 & \textcircled{3} & 0 & 3 \\ 0 & 0 & 0 & \textcircled{1} & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \\ u \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$x, y, z, t$  = pivot variables ;  $u$  = free variable

$$\boxed{t=0}$$

$$3z + 3u = 0$$

$$3z = -3u$$

$$\boxed{z = -u}$$

$$-2y + 4z - 3t + 2u = 0$$

$$-2y = -4z - 2u$$

$$+y = 2z + u$$

$$y = -2u + u$$

$$\boxed{y = -u}$$

$$x + y - 2z - u = 0$$

$$x = -y + 2z + u$$

$$x = u - 2u + u$$

$$\boxed{x=0}$$

$$\therefore N(A) = \begin{bmatrix} x \\ y \\ z \\ t \\ u \end{bmatrix} = \begin{bmatrix} 0 \\ -u \\ -u \\ 0 \\ u \end{bmatrix} = u \begin{bmatrix} 0 \\ -1 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{Basis} = \begin{bmatrix} 0 \\ -1 \\ -1 \\ 0 \\ 1 \end{bmatrix} ; \text{Dim}\{N(A)\} = 1$$

TO find  $N(AT)$  :-

(21)

$$A = \begin{bmatrix} 1 & 1 & -2 & 0 & -1 \\ 0 & -2 & 4 & -3 & 2 \\ 0 & 0 & 3 & 0 & 3 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & -2 & 0 & 0 \\ -2 & 4 & 3 & 0 \\ 0 & -3 & 0 & 1 \\ -1 & 2 & 3 & 0 \end{bmatrix}$$

$N(AT)$  is special solutions to  $A^T \cdot x = 0$

$$\begin{bmatrix} \textcircled{1} & 0 & 0 & 0 \\ 1 & \textcircled{-2} & 0 & 0 \\ -2 & 4 & \textcircled{3} & 0 \\ 0 & -3 & 0 & 1 \\ -1 & 2 & 3 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & -2 & 0 & 0 \\ -2 & 4 & 3 & 0 \\ 0 & -3 & 0 & 1 \\ -1 & 2 & 3 & 0 \end{bmatrix} \xrightarrow[\begin{smallmatrix} R_2 - R_1 \\ R_3 + 2R_1 \\ R_5 + R_1 \end{smallmatrix}]{\begin{smallmatrix} R_2 - R_1 \\ R_3 + 2R_1 \\ R_5 + R_1 \end{smallmatrix}} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 4 & 3 & 0 \\ 0 & -3 & 0 & 1 \\ 0 & 2 & 3 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 4 & 3 & 0 \\ 0 & -3 & 0 & 1 \\ 0 & 2 & 3 & 0 \end{bmatrix} \xrightarrow{R_3 + 2R_2} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & -3 & 0 & 1 \\ 0 & 2 & 3 & 0 \end{bmatrix}$$

$$-3 - 2x = 0$$

$$-3 = 2x$$

$$x = -3/2$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & -3 & 0 & 1 \\ 0 & 2 & 3 & 0 \end{bmatrix} \xrightarrow[\begin{smallmatrix} R_4 + 3/2 R_2 \\ R_5 + R_2 \end{smallmatrix}]{\begin{smallmatrix} R_4 + 3/2 R_2 \\ R_5 + R_2 \end{smallmatrix}} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 3 & 0 \end{bmatrix} \xrightarrow{R_5 - R_3} \begin{bmatrix} \textcircled{1} & 0 & 0 & 0 \\ 0 & \textcircled{-2} & 0 & 0 \\ 0 & 0 & \textcircled{3} & 0 \\ 0 & 0 & 0 & \textcircled{1} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(22)

$$\begin{bmatrix} \textcircled{1} & 0 & 0 & 0 \\ 0 & \textcircled{-2} & 0 & 0 \\ 0 & 0 & \textcircled{3} & 0 \\ 0 & 0 & 0 & \textcircled{1} \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$t=0; z=0; y=0; x=0$$

$N(A^T) = \text{zero vector in } \mathbb{R}^4.$

8 i) Which vectors  $(b_1, b_2, b_3)$  are in the column space of  $A$ ?  
Which combination of the rows of  $A$  give 0?

$$A = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 \end{bmatrix}$$

Solution:-  $\begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$$\begin{bmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 - 2R_1} \begin{bmatrix} \textcircled{1} & 2 & 0 & 3 \\ 0 & 0 & 0 & \textcircled{-2} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$C(A) = \left\{ c_1 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix} \mid c_1, c_2 \in \mathbb{R} \right\}$$

For  $b_2 = 0$  and for any  $b_1, b_3$  are in column space of  $A$ .

$$\begin{bmatrix} 1 & 2 & 0 & 3 & : & b_1 \\ 0 & 0 & 0 & 0 & : & b_2 \\ 2 & 4 & 0 & 1 & : & b_3 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 2 & 0 & 3 & : & b_1 \\ 2 & 4 & 0 & 1 & : & b_3 \\ 0 & 0 & 0 & 0 & : & b_2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 0 & 3 & : & b_1 \\ 2 & 4 & 0 & 1 & : & b_3 \\ 0 & 0 & 0 & 0 & : & b_2 \end{bmatrix} \xrightarrow{R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 0 & 3 & : & b_1 \\ 0 & 0 & 0 & -2 & : & b_3 - 2b_1 \\ 0 & 0 & 0 & 0 & : & b_2 \end{bmatrix}$$

NO combination of rows give zero.

15) Find left/right inverse (whichever possible) for (23)  
the following matrices :-

i)  $\begin{bmatrix} 3 & 0 & 1 \\ 1 & 0 & -2 \end{bmatrix}$  ii)  $\begin{bmatrix} 3 & 1 \\ -1 & 1 \\ 1 & -1 \end{bmatrix}$

Solution :- i)  $\begin{bmatrix} 3 & 0 & 1 \\ 1 & 0 & -2 \end{bmatrix}_{2 \times 3}$

Here  $\rho(A) = 2 = m$  ∴ Right inverse exists

∴ Best right inverse  $= A^T(A \cdot A^T)^{-1}$

$$A = \begin{bmatrix} 3 & 0 & 1 \\ 1 & 0 & -2 \end{bmatrix}; A^T = \begin{bmatrix} 3 & 1 \\ 0 & 0 \\ 1 & -2 \end{bmatrix}$$

$$A \cdot A^T = \begin{bmatrix} 3 & 0 & 1 \\ 1 & 0 & -2 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 0 & 0 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} 10 & 1 \\ 1 & 5 \end{bmatrix}$$

$$(A \cdot A^T)^{-1} = \frac{1}{49} \begin{bmatrix} 10 & -1 \\ -1 & 5 \end{bmatrix}$$

$$A^T(A \cdot A^T)^{-1} = \begin{bmatrix} 3 & 1 \\ 0 & 0 \\ 1 & -2 \end{bmatrix}_{3 \times 2} \begin{bmatrix} 10 & -1 \\ -1 & 5 \end{bmatrix}_{2 \times 2} \frac{1}{49} = \frac{1}{49} \begin{bmatrix} 29 & 2 \\ 0 & 0 \\ 12 & -11 \end{bmatrix}$$

ii)  $\begin{bmatrix} 3 & 1 \\ -1 & 1 \\ 1 & -1 \end{bmatrix}_{3 \times 2} \xrightarrow[\substack{R_2 + \frac{1}{3}R_1 \\ R_3 - \frac{1}{3}R_1}]{R_2 + \frac{1}{3}R_1} \begin{bmatrix} 3 & 1 \\ 0 & 4/3 \\ 0 & -4/3 \end{bmatrix} \xrightarrow{R_3 + R_2} \begin{bmatrix} 3 & 1 \\ 0 & 4/3 \\ 0 & 0 \end{bmatrix}$

$$1 + \frac{1}{3} \quad -1 - \frac{1}{3}$$

$\rho(A) = 2 = n$ ; ∴ left inverse exists

The Best left inverse  $= (A^T \cdot A)^{-1} \cdot A^T = \begin{bmatrix} 11 & 1 \\ 1 & 3 \end{bmatrix}$

$$A = \begin{bmatrix} 3 & 1 \\ -1 & 1 \\ 1 & -1 \end{bmatrix}; A^T = \begin{bmatrix} 3 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}; A^T \cdot A = \begin{bmatrix} 3 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ -1 & 1 \\ 1 & -1 \end{bmatrix}$$

(24)

$$A^T \cdot A = \begin{bmatrix} 11 & 1 \\ 1 & 3 \end{bmatrix}; (A^T \cdot A)^{-1} = \frac{1}{32} \begin{bmatrix} 11 & -1 \\ -1 & 3 \end{bmatrix}$$

$$\therefore \text{Best least inverse} = (A^T \cdot A)^{-1} \cdot A^T$$

$$= \frac{1}{32} \begin{bmatrix} 11 & -1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 3 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$$

$2 \times 2 \qquad \qquad \qquad 2 \times 3$

$$= \frac{1}{32} \begin{bmatrix} 32 & -12 & 12 \\ 0 & 4 & -4 \end{bmatrix}$$

$$(A^T \cdot A)^{-1} \cdot A^T = \begin{bmatrix} 1 & -3/8 & 3/8 \\ 0 & 1/8 & 1/8 \end{bmatrix}$$

10) Find a basis and dimension of the subspace  $W$  of

$$V = M_{2 \times 2} \text{ spanned by } A = \begin{bmatrix} 1 & -5 \\ -4 & 2 \end{bmatrix}, B = \begin{bmatrix} 1 & 1 \\ -1 & 5 \end{bmatrix} \quad C = \begin{bmatrix} 2 & -4 \\ -5 & 7 \end{bmatrix}$$

$$D = \begin{bmatrix} 1 & -7 \\ -5 & 1 \end{bmatrix}$$

Solution :

$$\begin{bmatrix} 1 & 1 & 2 & 1 \\ -4 & -1 & -5 & -5 \\ -5 & 1 & -4 & -7 \\ 2 & 5 & 7 & 1 \end{bmatrix} \xrightarrow{\substack{R_2 + 4R_1 \\ R_3 + 5R_1 \\ R_4 - 2R_1}} \begin{bmatrix} 1 & 1 & 2 & 1 \\ 0 & 3 & 3 & -1 \\ 0 & 6 & 6 & -2 \\ 0 & 3 & 3 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 2 & 1 \\ 0 & 3 & 3 & -1 \\ 0 & 6 & 6 & -2 \\ 0 & 3 & 3 & -1 \end{bmatrix} \xrightarrow{\substack{R_3 - 2R_2 \\ R_4 - R_2}} \begin{bmatrix} \textcircled{1} & 1 & 2 & 1 \\ 0 & \textcircled{3} & 3 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$C(A) = \left\{ c_1 \begin{bmatrix} 1 & -5 \\ -4 & 2 \end{bmatrix} + c_2 \begin{bmatrix} 1 & 1 \\ -1 & 5 \end{bmatrix} \mid c_1, c_2 \in \mathbb{R} \right\}$$

Basis of  $W$  :  $\begin{bmatrix} 1 & -5 \\ -4 & 2 \end{bmatrix}$  &  $\begin{bmatrix} 1 & 1 \\ -1 & 5 \end{bmatrix}$

Dimension of  $W$  : 2

9) If the set of vectors  $\{u, v, w\}$  are linearly independent vectors, then show that  $u+v, v+w, u+w$  are linearly independent?

Solution Since  $u, v$  &  $w$  are linearly independent

$$c_1 \cdot u + c_2 \cdot v + c_3 \cdot w = 0 \quad \text{--- (1)}$$

To show the vectors  $u+v, v+w$  &  $u+w$  are linearly independent :-

$$\alpha \{u+v\} + \beta \{v+w\} + \delta \{u+w\} = 0$$

$$\alpha \check{u} + \alpha \check{v} + \beta \check{v} + \beta \check{w} + \delta \check{u} + \delta \check{w} = 0$$

$$(\alpha + \delta)u + (\alpha + \beta)v + (\beta + \delta)w = 0 \quad \text{--- (2)}$$

We know that the vectors  $u, v$  &  $w$  are linearly independent  $\Rightarrow \alpha + \delta = 0; \alpha + \beta = 0; \beta + \delta = 0$

From (1) & (2)

$$\alpha + \delta = c_1 \Rightarrow \alpha + \delta = 0 \quad \text{--- (3)}$$

$$\alpha + \beta = c_2 \Rightarrow \alpha + \beta = 0 \quad \text{--- (4)}$$

$$\beta + \delta = c_3 \Rightarrow \beta + \delta = 0 \quad \text{--- (5)}$$

consider equations (3) & (4)

$$\alpha + \delta = 0$$

$$\alpha + \beta = 0$$

$$\begin{array}{r} (-) \quad (-) \\ \hline \end{array}$$

$$\delta - \beta = 0$$

$$\delta = \beta$$

$$\delta + \beta = 0$$

$$\beta + \beta = 0$$

$$2\beta = 0$$

$$\boxed{\beta = 0}$$

$$\boxed{\delta = 0}$$

$$\alpha + \delta = 0$$

$$\Rightarrow \boxed{\alpha = 0}$$

Since  $\alpha, \beta$  &  $\delta = 0 \Rightarrow$  vectors  $u+v, v+w$  &  $u+w$  are linearly independent.

(26)

7) Find the basis for the plane  $x - 2y + 2z = 0$  in  $\mathbb{R}^3$ . Then find a basis for the intersection of that plane with the  $xy$ -plane.

Solution

$$x - 2y + 2z = 0$$

$$x = 2y - 2z$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2y - 2z \\ y \\ z \end{pmatrix} = y \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + z \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}$$

o.o Basis vectors  $\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$

We need to find a basis for the intersection of the above plane with the  $xy$ -plane.

For  $xy$  plane,  $z = 0$

$$\text{o.o} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2y \\ y \\ 0 \end{pmatrix} = y \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$$

o.o Basis is a line in  $\mathbb{R}^3$ .

3) Find the dimensions and basis for each of the four fundamental subspaces of the following matrix :-

$$\begin{bmatrix} 1 & 2 & 1 & 2 \\ 1 & 2 & 1 & 3 \\ 3 & 6 & 3 & 7 \end{bmatrix}$$

Solution

$$\begin{bmatrix} 1 & 2 & 1 & 2 \\ 1 & 2 & 1 & 3 \\ 3 & 6 & 3 & 7 \end{bmatrix} \xrightarrow{\substack{R_2 - R_1 \\ R_3 - 3R_1}} \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_3 - R_2} \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$C(A) = \left\{ c_1 \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 3 \\ 7 \end{bmatrix} \mid c_1, c_2 \in \mathbb{R} \right\}$$

$$\dim\{C(A)\} = 2$$

$$C(A^T) = \left\{ c_1 \begin{bmatrix} 1 \\ 2 \\ 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 2 \\ 1 \\ 3 \end{bmatrix} \mid c_1, c_2 \in \mathbb{R} \right\}$$

(OR)

$$C(A^T) = \left\{ c_1 \begin{bmatrix} 1 \\ 2 \\ 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \mid c_1, c_2 \in \mathbb{R} \right\}$$

$$\dim\{C(A^T)\} = 2$$

(28)

TO find  $N(A)$  :-  $A \cdot x = 0$ 

$$\begin{bmatrix} \textcircled{1} & 2 & 1 & 2 \\ 0 & 0 & 0 & \textcircled{1} \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Pivot variables =  $x, t$   
Free variables =  $y, z$

$$t=0; \quad x+2y+z+2t=0 \Rightarrow x+2y+z=0$$

$$x = -2y - z$$

Let  $y=0$  &  $z=1; t=0; x=-1$

$$\therefore \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

Let  $y=1$  &  $z=0; t=0$  &  $x=-2$

$$\therefore \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} -2y - z \\ y \\ z \\ 0 \end{bmatrix} = y \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + z \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

Basis vectors :-  $\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}; \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$   $\dim\{N(A)\} = 2$

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 1 & 2 & 1 & 3 \\ 3 & 6 & 3 & 7 \end{bmatrix}; A^T = \begin{bmatrix} 1 & 1 & 3 \\ 2 & 2 & 6 \\ 1 & 1 & 3 \\ 2 & 3 & 7 \end{bmatrix} \xrightarrow{\substack{R_2 - 2R_1 \\ R_3 - R_1 \\ R_4 - 2R_1}} \begin{bmatrix} 1 & 1 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_4} \begin{bmatrix} \textcircled{1} & 1 & 3 \\ 0 & \textcircled{1} & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R} \begin{bmatrix} \textcircled{1} & 1 & 3 \\ 0 & \textcircled{1} & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} \textcircled{1} & 1 & 3 \\ 0 & \textcircled{1} & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$x, y$  = pivot variable;  $z$  = free variable

$$x + y + 3z = 0 \quad ; \quad y + z = 0 \quad ; \quad y = -z$$

$$x = -y - 3z \quad ; \quad x = z - 3z = -2z$$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2z \\ -z \\ z \end{bmatrix} = z \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix}$$

Basis vector

$$\text{Dimension } N\{A^T\} = 1$$

4) If the vectors  $(1, 1, 2)$ ,  $(1, 2, 4)$  and  $(2, 4, 8)$  span the column space of  $A$ , determine whether or not the vector  $b = (2, 3, 5)$  is in  $C(A)$ . What value should replace the third component "5" in vector  $b$  so the system  $Ax = b$  has infinitely many solutions?

Solution

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 4 \\ 2 & 4 & 8 \end{bmatrix} \xrightarrow{\substack{R_2 - R_1 \\ R_3 - 2R_1}} \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & 2 & 4 \end{bmatrix} \xrightarrow{R_3 - 2R_2} \begin{bmatrix} \textcircled{1} & 1 & 2 \\ 0 & \textcircled{1} & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

30) 
$$\begin{bmatrix} \textcircled{1} & 1 & 2 \\ 0 & \textcircled{1} & 2 \\ 0 & 0 & 0 \end{bmatrix}; C(A) = \left\{ c_1 \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} \mid c_1, c_2 \in \mathbb{R} \right\}$$

$b = (2, 3, 5)$  is not in  $C(A)$   $\because$  it is not a linear combination of  $\in(A)$ . If we have 6 instead of "5" in the third component, then  $b$  will be in  $C(A)$ .  $c_1 = 1$  &  $c_2 = 1$   $\because$   $1 \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 6 \end{bmatrix}$

2) Show that the vectors  $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 2 \\ 1 & -1 \end{pmatrix}$  generate the vector  $\begin{pmatrix} 3 & -1 \\ 1 & -2 \end{pmatrix}$  in the vector space of all  $2 \times 2$  matrices.

Solution:-

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \\ 0 & 1 & -1 \end{bmatrix} \xrightarrow[R_3 - R_1]{R_2 - R_1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 1 & -1 \end{bmatrix} \xrightarrow[R_4]{R_4 - R_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & -2 \end{bmatrix} \xrightarrow{R_4 + R_3} \begin{bmatrix} \textcircled{1} & 0 & 0 \\ 0 & \textcircled{1} & 1 \\ 0 & 0 & \textcircled{2} \\ 0 & 0 & 0 \end{bmatrix}$$

column space  $\therefore c_1 \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} + c_3 \begin{pmatrix} 0 & 2 \\ 1 & -1 \end{pmatrix}$

This should generate  $\begin{pmatrix} 3 & -1 \\ 1 & -2 \end{pmatrix}$   $\begin{matrix} c_1 = 3 \\ c_2 = 0 \\ c_3 = -2 \end{matrix}$

1) Let  $v_1 = (1, 2, 1)$ ,  $v_2 = (3, 1, 5)$  and  $v_3 = (3, -4, 7)$ . Show 31 that the subspaces spanned by  $\{v_1, v_2\}$  and  $\{v_1, v_2, v_3\}$  are the same.

Solution: consider only  $(v_1, v_2)$

$$\begin{bmatrix} 1 & 3 \\ 2 & 1 \\ 1 & 5 \end{bmatrix} \xrightarrow[R_3 - R_1]{R_2 - 2R_1} \begin{bmatrix} 1 & 3 \\ 0 & -5 \\ 0 & 2 \end{bmatrix} \xrightarrow{R_3 + \frac{2}{5}R_2} \begin{bmatrix} 1 & 3 \\ 0 & -5 \\ 0 & 0 \end{bmatrix}$$

$$2 + 5x = 0$$

$$2 = -5x$$

$$x = -\frac{2}{5}$$

$$C(A) = \left\{ c_1 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 3 \\ 1 \\ 5 \end{bmatrix} ; c_1, c_2 \in \mathbb{R} \right\}$$

Consider  $v_1, v_2, v_3$

$$\begin{bmatrix} 1 & 3 & 3 \\ 2 & 1 & -4 \\ 1 & 5 & 7 \end{bmatrix} \xrightarrow[R_3 - R_1]{R_2 - 2R_1} \begin{bmatrix} 1 & 3 & 3 \\ 0 & -5 & -10 \\ 0 & 2 & 4 \end{bmatrix} \xrightarrow{R_3 + \frac{2}{5}R_2} \begin{bmatrix} 1 & 3 & 3 \\ 0 & -5 & -10 \\ 0 & 0 & 0 \end{bmatrix}$$

$$2 + 5x = 0 \Rightarrow x = -\frac{2}{5}$$

$$4 + \frac{2}{5}(-10)$$

$$C(A) = \left\{ c_1 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 3 \\ 1 \\ 5 \end{bmatrix} \mid c_1, c_2 \in \mathbb{R} \right\}$$

Both have same column space.  $\therefore$  the subspaces spanned by  $\{v_1, v_2\}$  and  $\{v_1, v_2, v_3\}$  are the same.